

Macaque OLS and lag-embedding audit

210 single-channel motor-LFP recordings | 2 macaques

Monkey M: 115 recordings / 48 days. Monkey T: 95 recordings / 30 days.

Balanced short-delay trials; last 500 ms before GO; saved five outer folds.

MAIN FINDINGS

AR prediction: mean held-out 10-ms NMSE is 0.00275 for M and 0.00249 for T. The corresponding mean R-squared is approximately 0.997 for both animals.

AR order: fold-wise median $p=18$ for M and $p=16$ for T. The most frequent recording-modal order is $p=19$ for M and $p=14$ for T. All fold choices are saved.

Embedding selection: every fold chooses $m=9$, $\tau=1$ ms. This reaches the upper dimension and lower delay limits of the tested grid; it is NOT an estimate of a nine-dimensional neural manifold or of the number of modes.

INTERPRETATION

These are offline predictions of 2-55-Hz, zero-phase-filtered, normalized signals sampled at 1 kHz. Whole-epoch preprocessing uses later samples within the pre-GO epoch. These results are not causal online forecasting.

Strong short-horizon predictability does not establish valid torus geometry. Selected design condition numbers are large (median about $1.31e8$ for AR). Numerically full rank does not make individual coefficients/poles reliable.

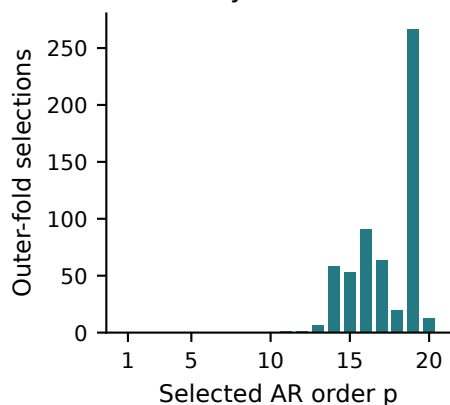
VALIDATION AND OUTPUTS

All 1,050 outer folds independently checked. Prior outputs unchanged. No PCA, geometric fitting, shape matrices, PINN, or decoding was run.

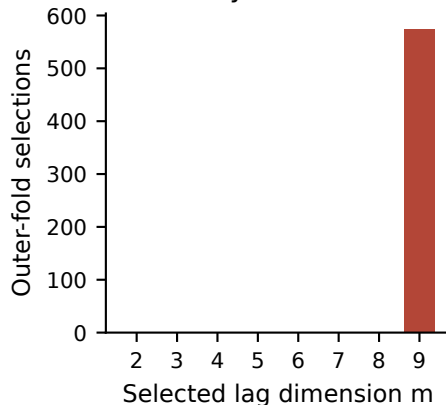
This booklet is a preview. Every figure is also available as a separate PDF, SVG and PNG with a CSV and caption. Each of the 210 recordings has its own order-sweep and embedding-sweep pair in plots/recordings/.

See README.md for full methods, warnings, metric definitions and tables. Folder: outputs/prego_ols_audit_v1

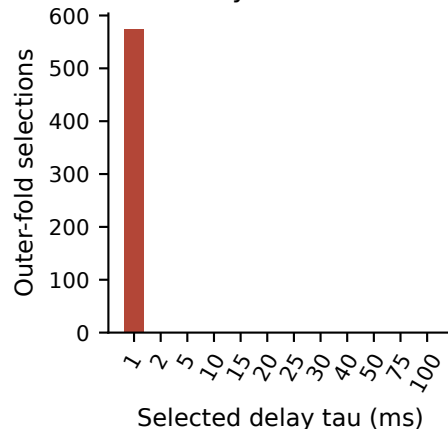
Monkey M: 115 LFPs



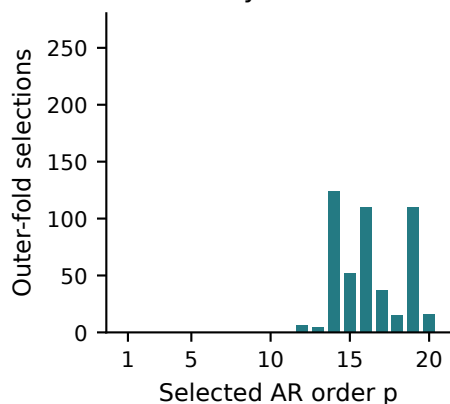
Monkey M: 115 LFPs



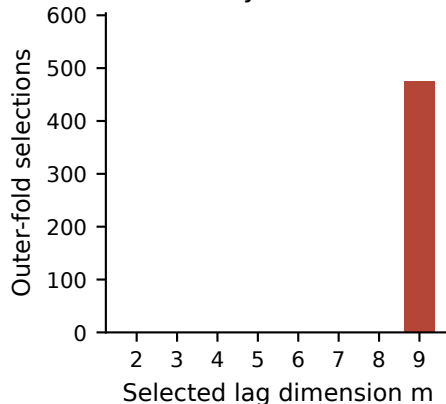
Monkey M: 115 LFPs



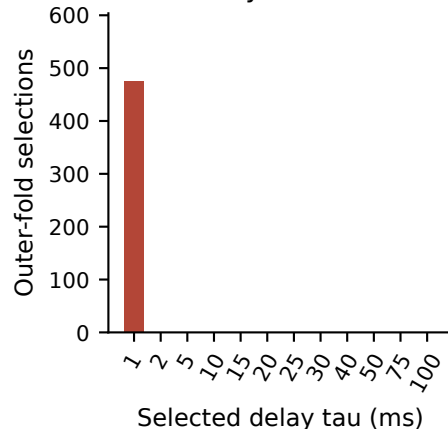
Monkey T: 95 LFPs



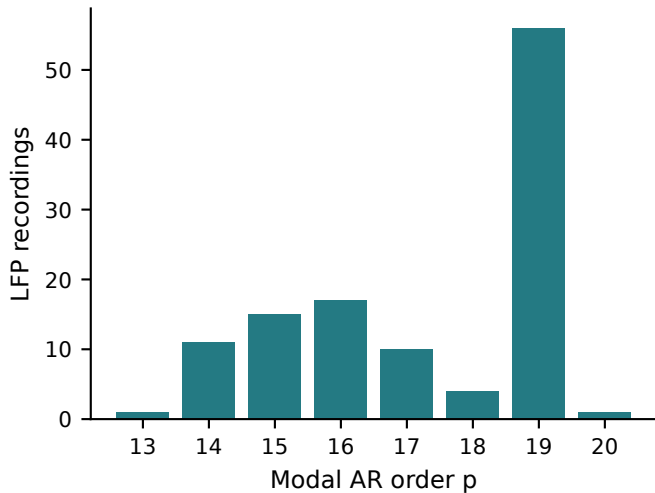
Monkey T: 95 LFPs



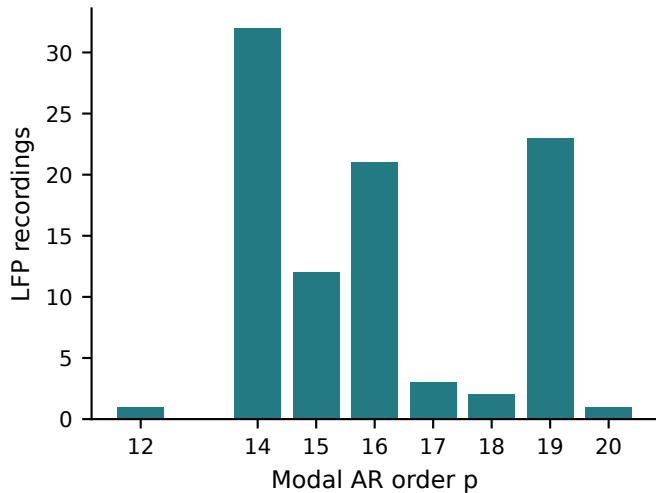
Monkey T: 95 LFPs

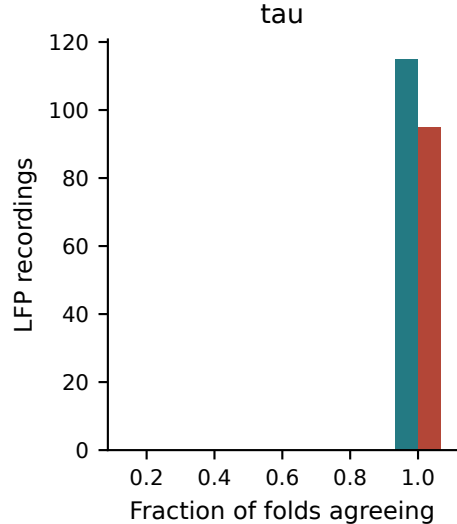
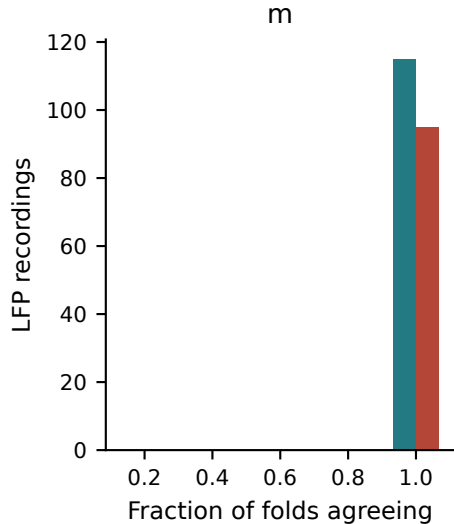
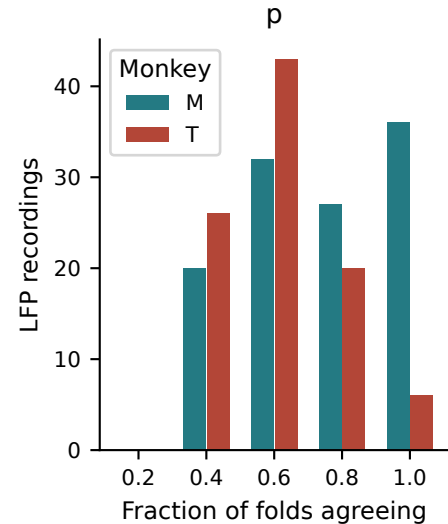


Monkey M



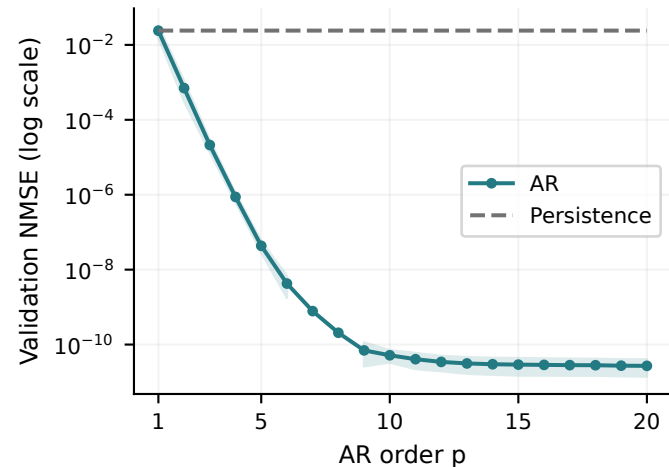
Monkey T



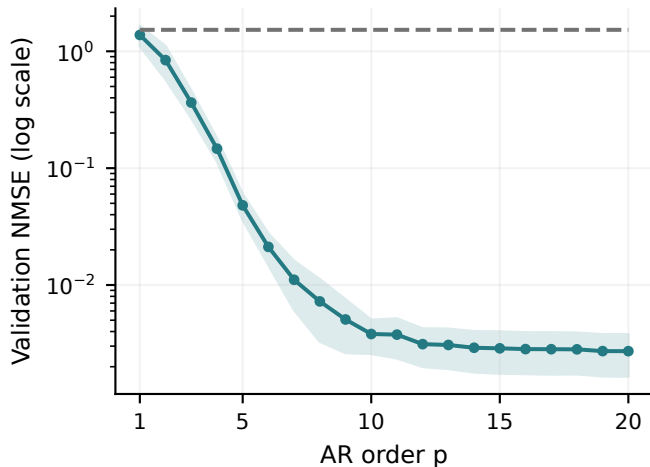


Monkey M: 115 LFP recordings

1-ms one-step

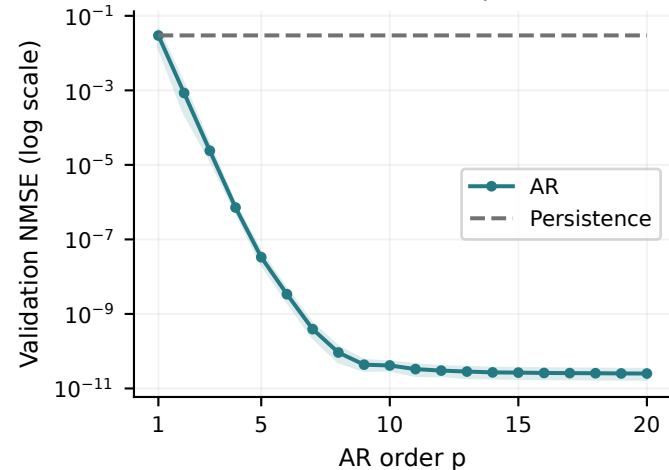


10-ms recursive

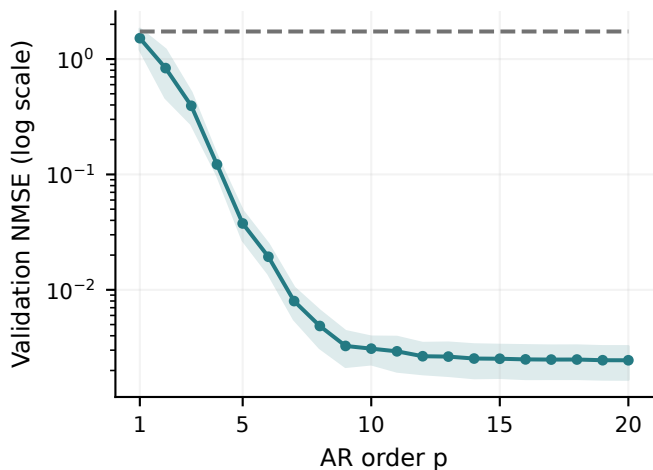


Monkey T: 95 LFP recordings

1-ms one-step

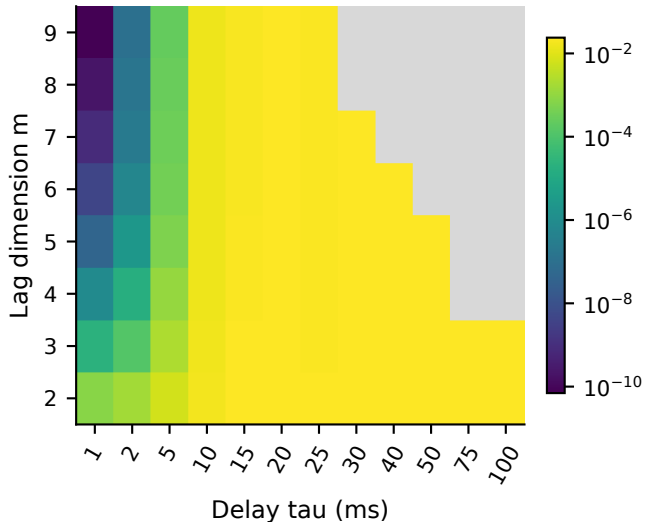


10-ms recursive

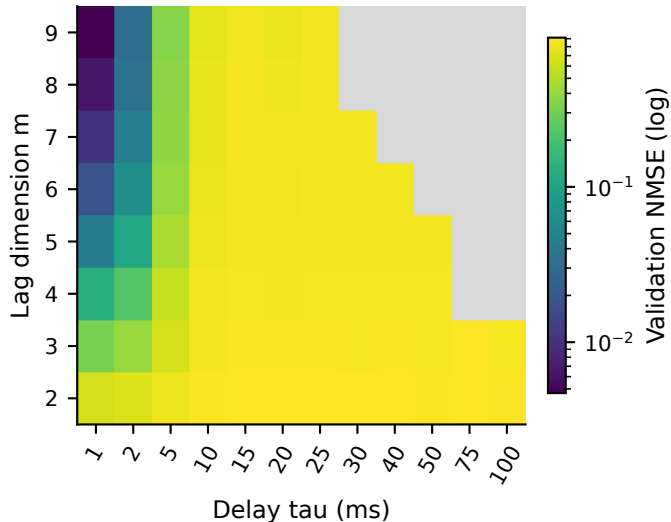


Monkey M: equal-recording averages

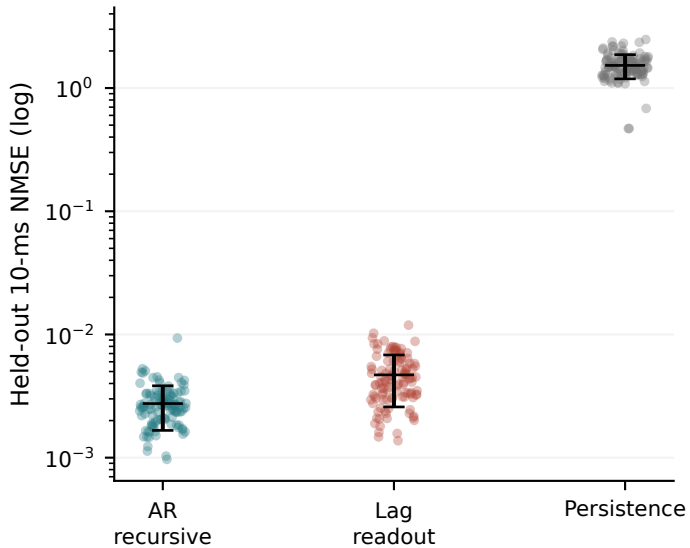
1-ms direct prediction



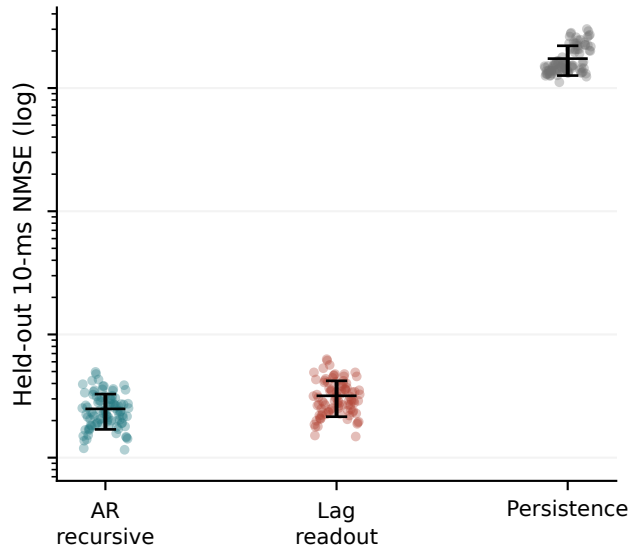
10-ms direct prediction



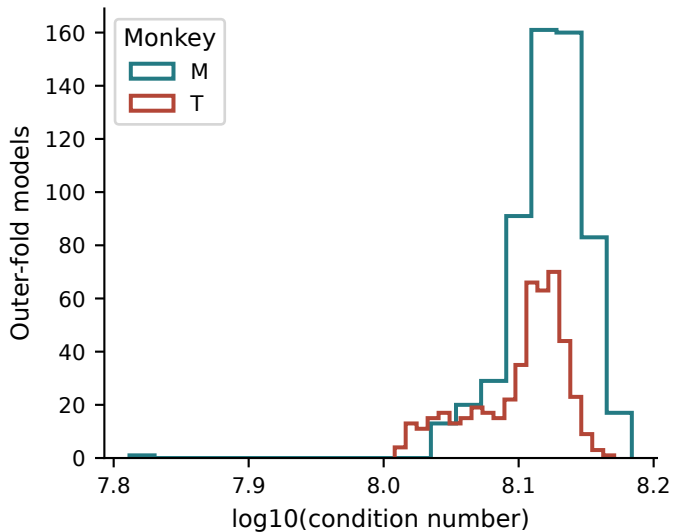
Monkey M



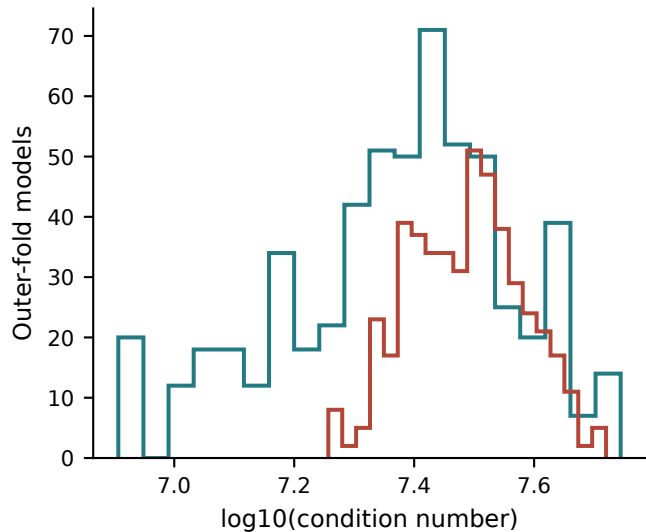
Monkey T



AR

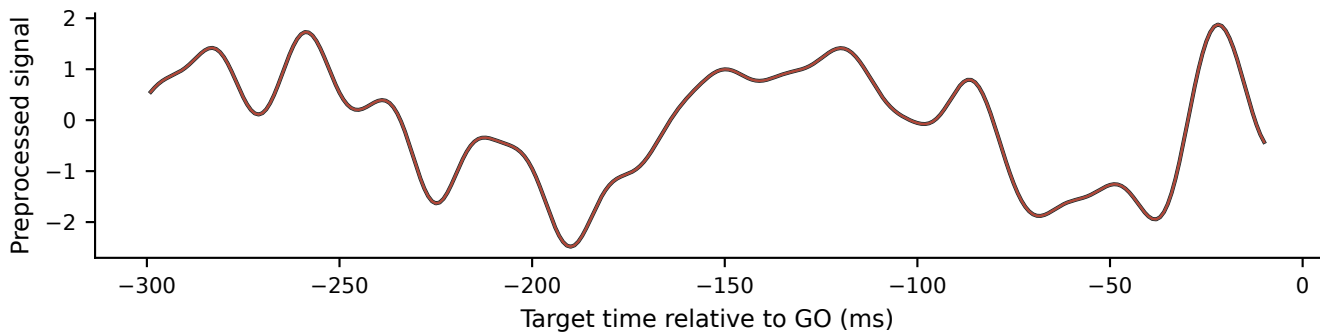


Lag readout

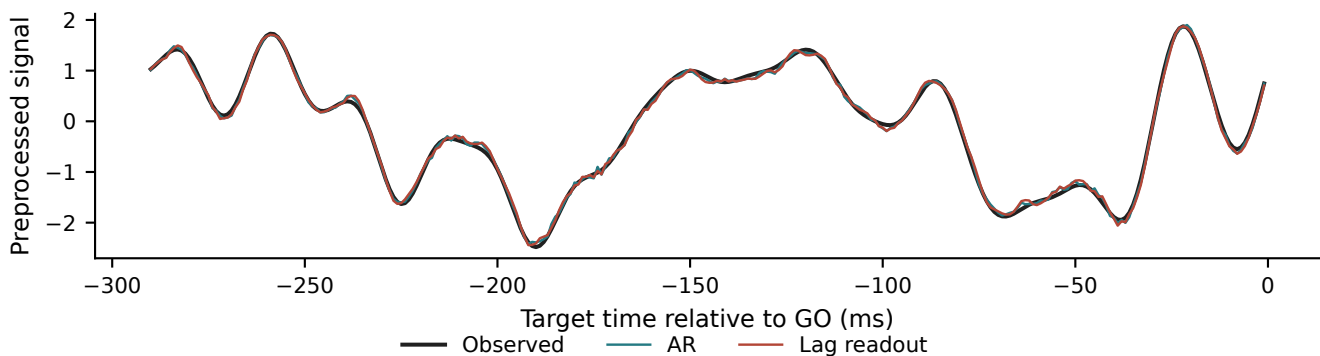


Monkey M | $p=17$, $m=9$, $\tau=1$ ms

1-ms horizon

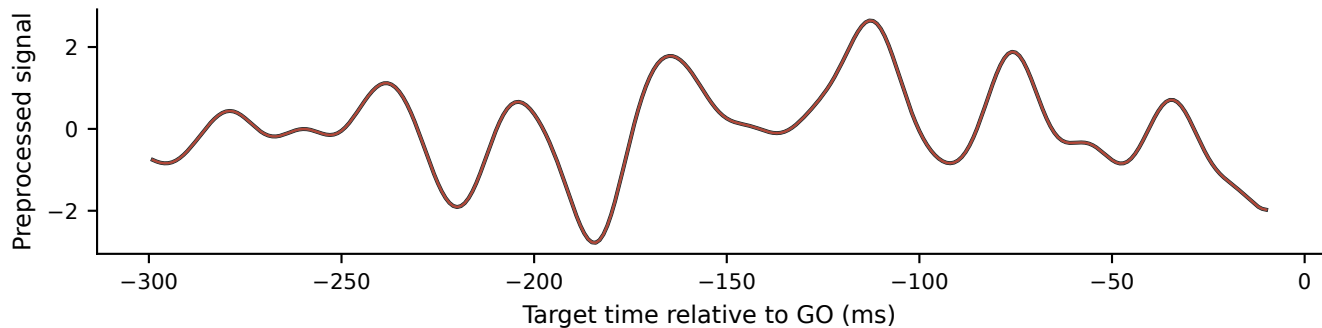


10-ms horizon

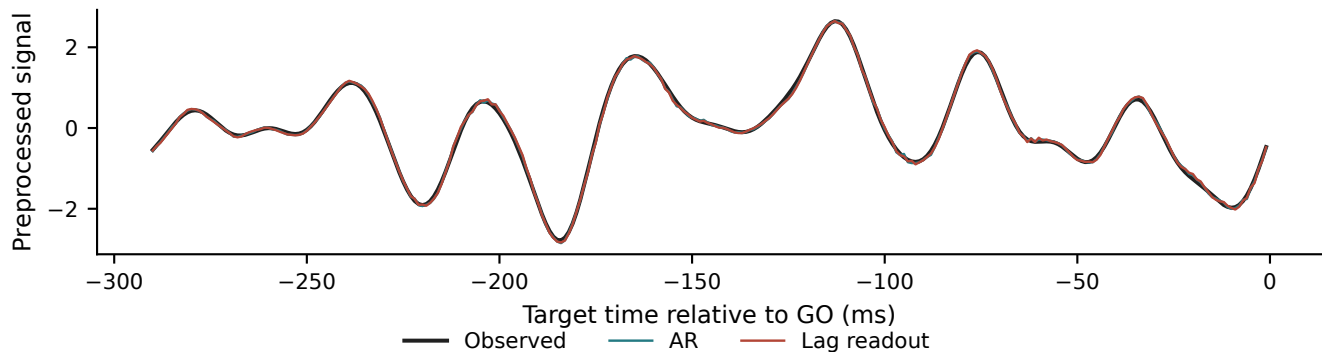


Monkey T | $p=16$, $m=9$, $\tau=1$ ms

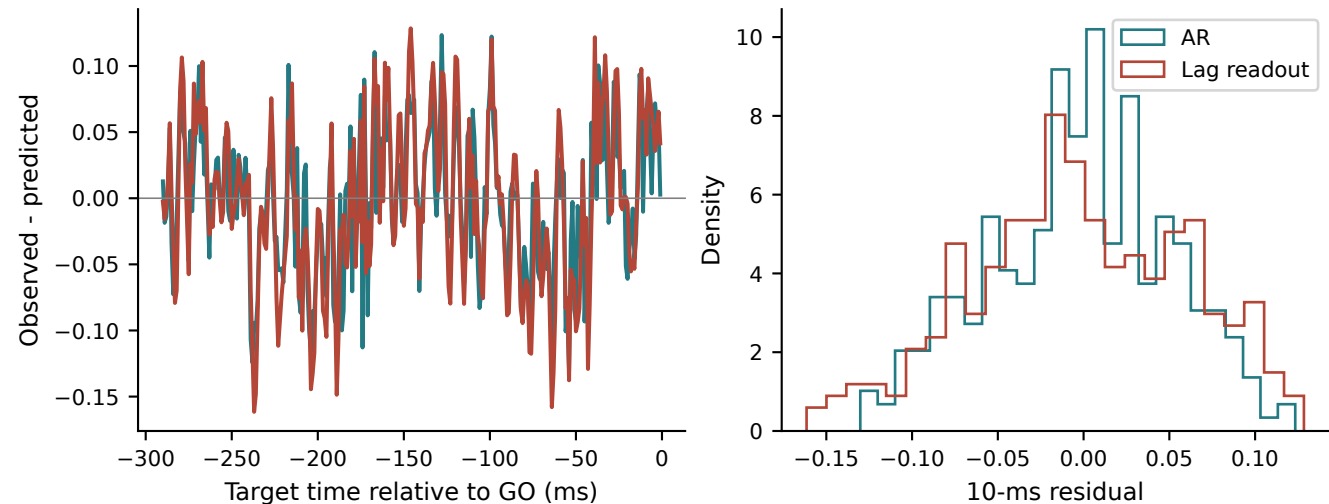
1-ms horizon



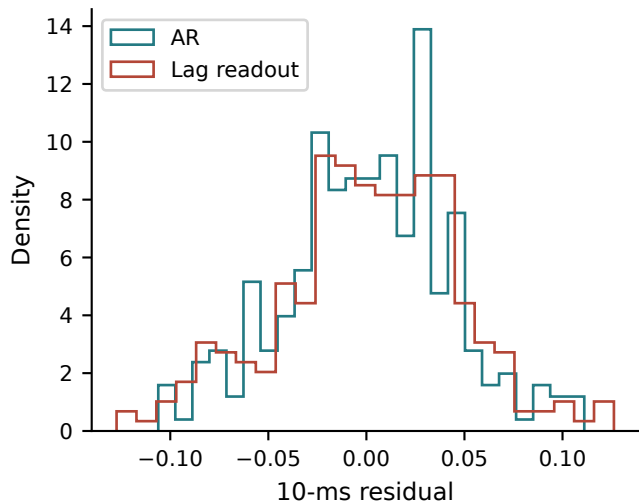
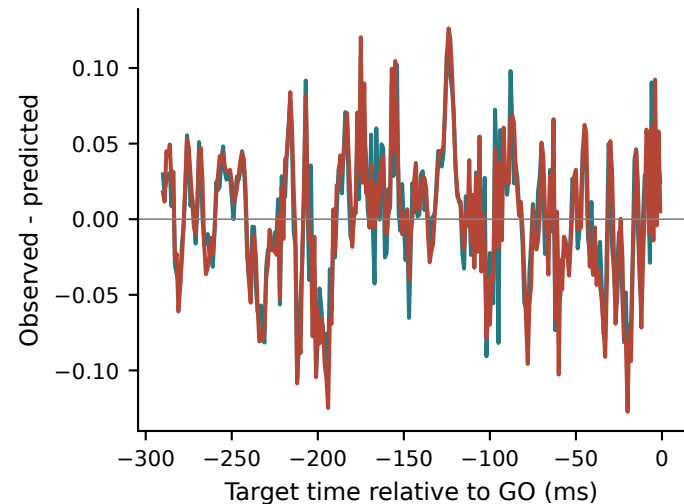
10-ms horizon



Monkey M | $p=17$, $m=9$, $\tau=1$ ms

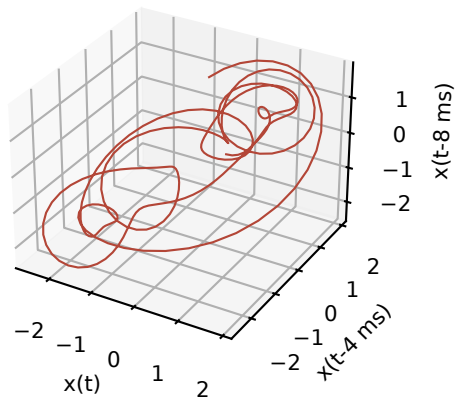


Monkey T | $p=16$, $m=9$, $\tau=1$ ms



Monkey M | $m=9$, $\tau=1$ ms

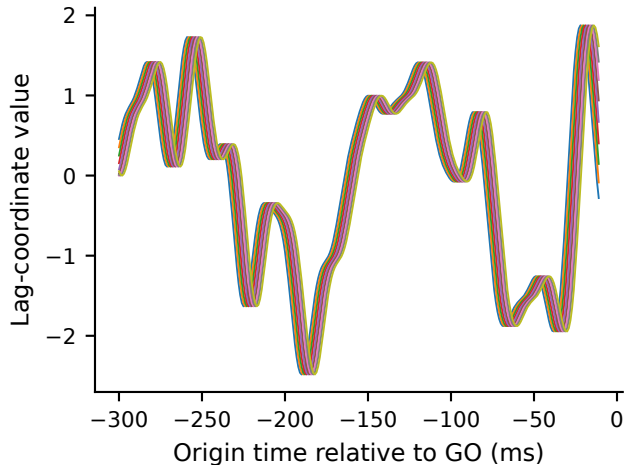
Coordinate subset (no PCA)



Lag (ms)

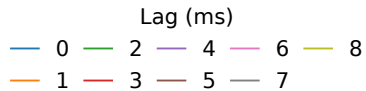
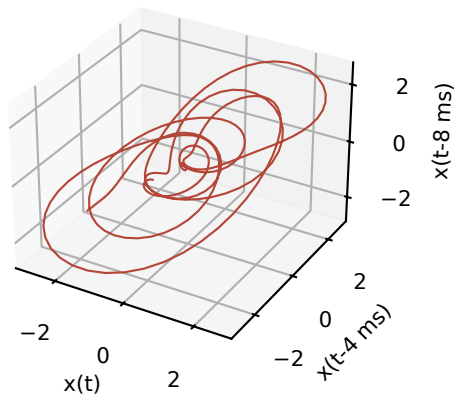
0	2	4	6	8
1	3	5	7	

All 9 coordinates



Monkey T | $m=9$, $\tau=1$ ms

Coordinate subset (no PCA)



All 9 coordinates

